◆ PRODUCT REQUIREMENTS DOCUMENT

PMOS Companion

Polyendocrine Metabolic Ovarian Syndrome

> The lifelong companion for the 1 in 8 women medicine renamed a condition for, and still does not treat.

TEAM

War Room

OWNERS

Dhodraj, Belson, Rahul, Satheesh Kumar

VERSION

1.0, July 2026

LAUNCH VEHICLE

Tech for Good 2026, GDG Coimbatore

DISTRIBUTION

mycyclesafe.ai, 500,000+ users

102 functional, 48 non-functional and 14 regulatory requirements

Single source of truth

PMOS Companion

Product Requirements Document

Version 1.0, July 2026. Single source of truth.

FIELD	VALUE
Team	War Room
Owners	Dhodraj, Belson, Rahul, Satheesh Kumar
Launch vehicle	Tech for Good 2026, GDG Coimbatore, 8 to 9 August
Distribution	mycyclesafe.ai (500,000+ users, 120 countries)

This document supersedes the product definition, build spec, and complete-features spec. Where those conflict with this one, this one governs.

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1. Executive summary

A daily well-being companion for women with PMOS. It treats an irregular cycle as signal rather than error, learns each user's personal cause-and-effect patterns, measures vitals from a bare phone camera, and produces evidence a doctor accepts in a seven-minute consultation. It runs on a ₹10,000 phone and requires no purchased hardware.

1.1 Why now

Three things converged inside eighteen months.

DATE	WHAT HAPPENED
2023	International Guideline establishes the metabolic framing
May 2026	**Monash-led global consensus renames PCOS to PMOS, published in The Lancet.** An 11-year global consensus involving 22,000 people, explicitly rejecting the cysts-and-fertility framing
Jul 2026	NICE draft guideline adopts PMOS. Consultation open 1 July to 11 August
Aug 2026	Tech for Good hackathon. We build
Dec 2026	NICE publishes. Three-year global rollout begins

The rename carries the entire thesis. Medicine spent eleven years concluding that this condition was never about cysts or fertility. It is a lifelong metabolic-endocrine condition. Every product on the market still treats it the old way, and NICE has already adopted the new name in its draft guideline. We build for the new definition.

1.2 What holds up under scrutiny

Six positions survived a competitive audit against products that already ship.

#	POSITION	WHY IT HOLDS
1	Experiment Mode	Competitors surface patterns. None reach a verdict. Pre-declared outcomes, washout periods, and honest results including "no detectable change"
2	Camera vitals, no hardware	No PMOS competitor captures any objective physiological signal
3	Clinical safety depth	Mental-health crisis routing, OSA screen-and-refer, guideline screening ledger. Absent across the category
4	India-native	Vernacular, Indian cuisine, Tele-MANAS, rupee pricing. The closest competitor ships iPhone-only
5	Zero-commerce evidence grading	We can grade evidence honestly because we never sell the supplements. Rivals funded by product sales cannot copy this

#	POSITION	WHY IT HOLDS
6	The mycyclesafe funnel	500k users, and the irregular-cycle segment is by definition this audience

2. The problem

2.1 The care pathway loses her at every stage

This is a loop rather than a funnel. She keeps returning to dismissal, and nothing accumulates.

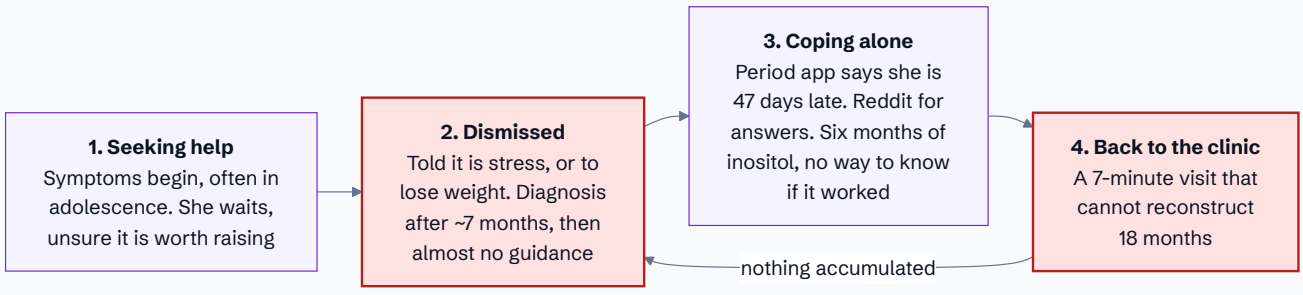


FIGURE 1

2.2 Evidence

PROBLEM	EVIDENCE	SEVERITY
Diagnostic delay	~7 months in India. Focus groups report 2 months to 13 years	Critical
Fertility-only framing	Clinicians emphasise weight loss and future fertility. Teenagers need help with acne, anxiety and cycles now	Critical
Mental-health burden	Anxiety ~6x and depression ~4x general population. 75% and 58% elevated scores	Critical
Body-image distress	84.5% score pathologically on appearance evaluation. Marriage-prospect pressure amplifies this in India	Critical
Anovulatory cycles	Gaps of 60 to 200+ days. Spotting confused with periods. Medication bleeds counted as cycles	Critical
Endometrial protection gap	Gaps over 90 days or fewer than 4 bleeds a year, with users unaware of the clinical significance	Critical
Sleep apnoea undetected	37% prevalence in PMOS against 6% general population. No app screens for it	High

PROBLEM	EVIDENCE	SEVERITY
Generic advice for an n-of-1 condition	What works differs substantially between individuals, and nothing personalises	Critical
No doctor-ready output	Users maintain spreadsheets by hand. A seven-minute consultation cannot reconstruct eighteen months	Critical
Tracker burnout	80-field symptom logs. Users quit by week three	High
Hardware paywalls	Metabolic insight sits behind a ₹4,000 wearable or a ₹8,400 monthly CGM	High

2.3 What she asks for

Across communities and app reviews, one request recurs almost verbatim: help me identify what personally influences my wellbeing, by logging periods, skin, mood, food, sleep and supplements to find correlations.

Nobody builds it. That sentence is the product.

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3. Goals and success metrics

3.1 Goals

ID	GOAL
G1	Make her irregular cycle legible, never framed as failure
G2	Tell her something true about her own body that nobody has told her
G3	Convert eighteen months of lived experience into evidence a doctor accepts in seven minutes
G4	Deliver objective physiological signal with no purchased hardware
G5	Cause no harm: no shame, no false precision, no missed crisis

3.2 Non-goals

Diagnosing PMOS. Replacing a clinician. Fertility and conception as a primary use case. Weight loss as an outcome. Selling supplements.

3.3 Metrics

METRIC	MVP	V1	V2
D7 retention	>40%	>50%	>55%
D90 retention	not measured	>25%	>35%
Weekly active loggers	>60% of MAU	>70%	>75%
Pattern reveals viewed	>1 per user per week	>1.5	not set
Doctor exports	not measured	>1 per user per quarter	>2
Experiment completion	not measured	>30% of started	>40%
Fingertip PPG success rate	>85%	>88%	>90%
Camera check-in adoption	not measured	>15% of DAU	>20%
Crisis-path integrity	100% scripted	100%	100%
NPS in beta	>50	>60	>65

Counter-metrics, where a rise means we are failing: uninstalls within 48 hours of a pattern reveal, reported distress following a skin capture, streak-related complaints, and any crisis path served by generated text.

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4. Users

Primary users are Indian women aged 18 to 35 with PMOS, diagnosed or suspecting it, phone-first, poorly served by generic apps and dismissive care. The secondary user is her clinician, who receives exports but never uses the product directly in v1.

PERSONA	SITUATION	WHAT SHE NEEDS
Priya, 26, Bengaluru	Diagnosed two years ago. Cycles run 30 to 60 days. Her doctor said to lose weight. Takes inositol without knowing whether it helps	Does this actually work for me?
Ananya, 21, Coimbatore	Undiagnosed. Acne, hair loss, irregular cycles. Told it is normal for her age. Deleted Flo after repeated pregnancy scares	Evidence to be taken seriously
Meera, 34, tier-2 city	Diagnosed eight years ago. Fatigue, poor sleep, snoring. Never screened for sleep apnoea or lipids. Sees four doctors who never coordinate	Nothing falls through

Users under 18 are excluded at launch. REG-001 explains why, and treats it as a cost we are choosing to bear, not a design preference.

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5. Competitive position

5.1 What competitors actually ship

PRODUCT	STATUS	WHAT IT ACTUALLY DOES
PMOS Vera	Shipped	iPhone-only, $5.99/month. Irregular-native framing, lab tracking, photo journal, doctor PDF, weight-neutral. No servers at all, so its privacy position is stronger than ours
Josie	Waitlist	Claims specialist prep reports, evidence-pack mode, four cycle states. Not shipped
Flo, Clue	Shipped	28-day assumption, the "you are late" harm, no metabolic layer
Veera, Gynoveda	Shipped	Clinic coaching or supplement commerce. Episodic, fertility-leaning

5.2 Where each product sits

PRODUCT	PMOS-NATIVE DEPTH	REACHES A VERDICT	POSITION
Flo, Clue	Low	No	Commodity tracker
Veera, Gynoveda	Medium	No	Coaching, no cycle intelligence
PMOS Vera	High	No	Depth without conclusion
Josie (unshipped)	High	Partial	Depth without conclusion
PMOS Companion	**High**	**Yes**	**The abandoned middle**

5.3 Parity against advantage

Internally we should be honest about which is which.

CAPABILITY	POSITION
Irregular-native, doctor export, weight-neutral, photo journal, lab tracking, specialist reports	Parity. PMOS Vera and Josie already claim these
Privacy architecture	Behind. Vera runs with zero servers
Experiment Mode, camera vitals, crisis routing, OSA screening, screening ledger, vernacular, zero-commerce, mycyclesafe funnel	Advantage

Two phrases in earlier drafts, a "Day 47" cycle framing and the label "Building My Case", appear near-verbatim in competitor marketing. Both need rewriting before anything ships publicly.

6. Architecture decisions

ADR-001: Capacitor shell with native plugins

The sensing core cannot run in a pure PWA. No web API exists for HealthKit or Health Connect, `getUserMedia` is unreliable inside WKWebView, and a documented WebKit defect blanks the camera preview when the torch is enabled.

We ship a Capacitor shell wrapping the web application, with native plugins for every sensing capability. The reasoning matters: a Capacitor native plugin runs the same Swift and Kotlin code a fully native app would run. For the PPG pipeline there is no capability or performance difference, because the plugin is native code. Going fully native changes only what renders the interface, at the cost of two codebases for a four-person team, and discards both the team's React experience and the existing mycyclesafe web assets.

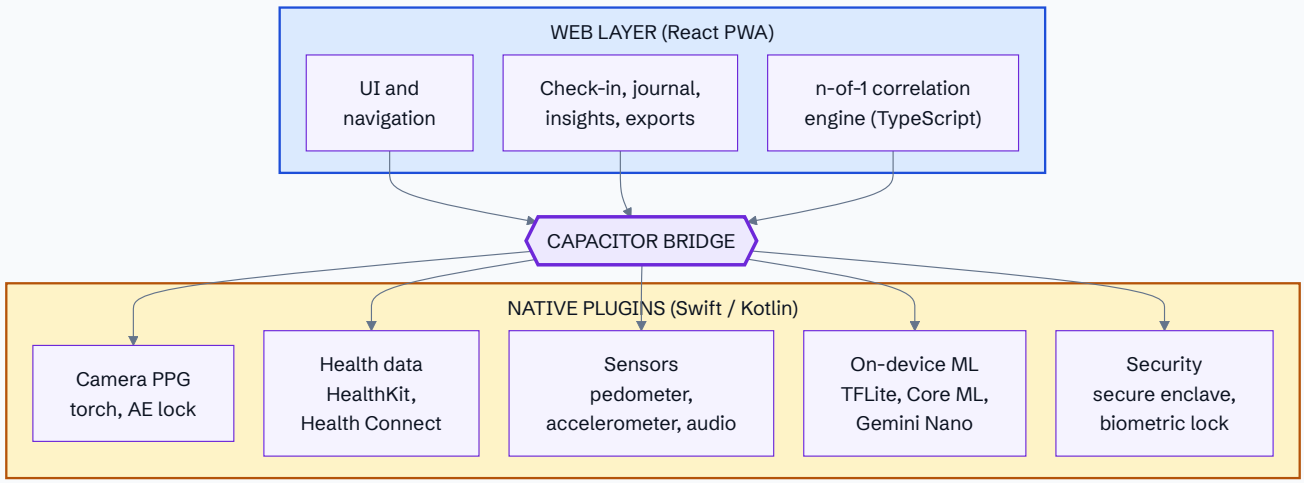


FIGURE 2

Roughly 85% of the product lives in the web layer. Only the sensing capabilities cross the bridge.

Consequences: one codebase, React skills retained, native performance where it matters, any screen replaceable with native later, and store distribution which health-data permissions require regardless.

Rejected alternatives. A pure PWA cannot deliver the sensing core. React Native is workable, and `react-native-vision-camera` frame processors suit PPG well, but it requires rebuilding the interface layer. Separate Swift and Kotlin codebases buy nothing that native plugins do not already provide.

ADR-002: Camera capture is always foreground and user-initiated

PLATFORM	BLOCKER
iOS	No unlock event API exists. `AVCaptureSession` interrupts with `videoDeviceNotAvailableInBackground` when the app backgrounds. The green recording indicator cannot be suppressed
Android	Background camera blocked since API 28. Android 14 and later throw `SecurityException` when a camera foreground service starts from the background, because `CAMERA` is a while-in-use permission
Web	More restricted than either

Google's PHRM works because Google owns the operating system's face-unlock hook. No PHRM SDK, API, ML Kit feature or Health Connect type exists, and the released model is licensed for non-commercial, non-clinical research under IRB approval, so we cannot use it. Every commercial rPPG SDK on the market (Binah.ai, NuraLogix, VitalLens) requires a foreground, user-held, still session, which is the strongest available evidence that no passive path exists for third parties.

The interaction model follows: notification, tap, foreground capture screen, 30 to 60 seconds, result.

ADR-003: No external devices

The claim: works on a ₹10,000 phone. No watch, no sensor, no strips, no cuff, nothing to buy.

REMOVED	REASON
CGM sync (Dexcom Stelo, Abbott Libre)	Stelo is not sold in India. Libre publishes no third-party API, so every working integration violates the LibreView terms. Cost runs about ₹8,400 a month, roughly half average per-capita monthly income. The 2023 Guideline names the OGTT at ₹100 to ₹300 as the accurate glycaemic test
Wearable dependency	India's wearables market contracted 4.0% in 2025, with smartwatches down 17.6%. Budget brands holding over half the market often ship no Health Connect integration
Glucometer, BP cuff	Recurring consumable cost, poor adherence

Laboratory tests are not a device dependency. They are a one-off service, mandated by the 2023 Guideline whether or not this product exists, and inexpensive in India at roughly ₹800 for fasting insulin. The constraint that follows: no insight, pattern, verdict, coaching action or export may be gated behind a laboratory value.

ADR-004: On-device inference as the privacy position

PMOS Vera runs with no servers. We cannot claim stronger privacy while sending data to a cloud model, so inference moves onto the device.

LAYER	LOCATION
PPG signal processing	On device, native
Skin scoring	On device, TFLite or Core ML

LAYER	LOCATION
Sleep audio analysis	On device, raw audio discarded
n-of-1 correlation engine	On device
Companion question answering	On-device LLM where hardware allows, cloud RAG fallback
Encrypted sync	Derived numbers only, never raw video, audio or images

7. Scope

IN SCOPE	OUT OF SCOPE
Irregular-native cycle engine	Diagnosis or condition-likelihood scores
30-second daily check-in	Blood oxygen, blood pressure, glucose
n-of-1 pattern engine	Any purchased hardware
Experiment Mode	Users under 18
Camera vitals: PPG and respiratory rate	Supplement sales
Doctor exports	Passive background camera
Mental health and crisis routing	Fertility-first funnel
Screening ledger and lab vault	Calorie or macro counting
Evidence-graded coaching	Facial emotion inference
On-device AI companion	Voice mood biomarkers

8. What we removed and why

A permanent record, so these do not quietly return.

FEATURE	VERDICT	REASON
Passive heart rate on unlock	Platform-blocked	An OS-vendor privilege. Blocked on iOS, Android and web
Blood oxygen from camera	Banned and infeasible	App Store guideline 1.4.1 forbids it explicitly. Phone sensors lack the 660nm and 940nm wavelengths oximetry requires
Blood pressure via pulse transit time	Banned and invalid	Same guideline. Uncalibrated cuffless BP is not valid, and renaming the output to "BP trend" does not avoid classification
Overnight ballistocardiography	Research-only	Published evidence covers dedicated under-mattress hardware, not a phone on a mattress
Facial action unit stress inference	Construct-invalid	Barrett et al. 2019 rejects reading internal states from facial movement
Voice mood biomarkers	Research-only	Cross-lingual transfer collapses to roughly 49%, which is chance on a binary task
Automated hirsutism scoring	Research-only	No validated automated system exists, and the reference standard itself carries roughly 14 points of interobserver variance
CGM sync and glucose meal tagging	Hardware dependency	See ADR-003
External keystroke capture	Platform-blocked	No iOS API. Android requires accessibility services, which fails store review
Google Fit	Retired	Sign-ups closed May 2024, support ends late 2026. Replaced by Health Connect

9. Functional requirements

Priorities: P0 must-have, P1 high, P2 differentiator, P3 future. Each requirement is written to be independently testable.

9.1 Cycle engine, FR-100

ID	REQUIREMENT	PHASE	PRI
FR-101	Estimate cycle windows probabilistically with confidence bands. The system SHALL NOT display a single predicted due-date	MVP	P0
FR-102	Present elapsed time as a day count relative to her own historical range. Copy SHALL NOT contain "late", "overdue", "missed", or pregnancy-scare language	MVP	P0
FR-103	Support named cycle states: regular, irregular, oligomenorrhea, amenorrhea, medically regulated, post-pill recovery. Each is valid, never an error	MVP	P0

ID	REQUIREMENT	PHASE	PRI
FR-104	Distinguish period, spotting, withdrawal bleed and breakthrough bleeding, with flow intensity, clots and pain association	MVP	P0
FR-105	Maintain a rolling cycle-interval distribution and a bleeds-per-year counter	MVP	P0
FR-106	Raise a calm escalation at gaps over 90 days or fewer than 4 bleeds a year, with endometrial-protection education and a one-tap gynaecology export. SHALL NOT suggest any specific medication	MVP	P0
FR-107	Estimate cycle phase probabilistically with visible uncertainty. SHALL NOT present hard ovulation prediction	MVP	P1
FR-108	Track medication-induced bleeds separately so interval arithmetic stays correct	v1	P1
FR-109	Surface cycle-variance reduction as the progress metric, not regularity against a population norm	v1	P1

9.2 Daily check-in, FR-200

ID	REQUIREMENT	PHASE	PRI
FR-201	Complete a full check-in in 30 seconds median, covering bleeding, sleep timing, movement, mood, energy, skin, cravings, pain and medications	MVP	P0
FR-202	Cap the symptom set at roughly 15 curated dimensions. SHALL NOT exceed 20 without an explicit scope decision	MVP	P0
FR-203	Provide one-tap severity entry and a "same as yesterday" shortcut	MVP	P1
FR-204	Provide pause mode as a first-class state. Returning SHALL NOT show guilt copy, broken streaks or backfill prompts	MVP	P0
FR-205	Detect obsessive tracking patterns and reduce prompting in response	v1	P1
FR-206	Provide a home-screen widget for five-second logging	v1	P1
FR-207	Provide the illustrated nine-area mFG hirsutism scale as a self-completed questionnaire, never automated	v1	P1
FR-208	Support voice-note journaling with on-device transcription into the symptom log. Voice is an input method, never a mood biomarker	v2	P2

9.3 Camera vitals, FR-300

All capture is foreground and user-initiated, per ADR-002.

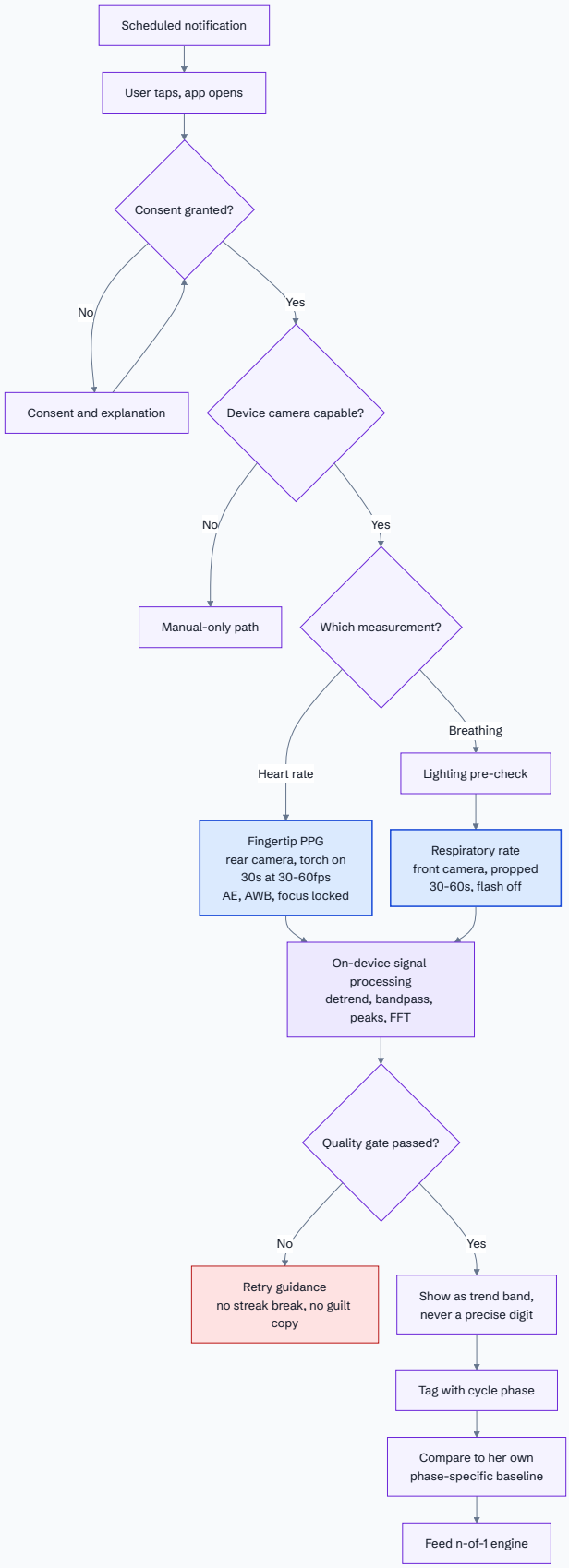

Scheduled notification
User taps, app opens
Consent granted?
No
Yes
Consent and explanation
Device camera capable?
No
Yes
Manual-only path
Which measurement?
Heart rate
Breathing
Lighting pre-check
Fingertip PPG
rear camera, torch on
30s at 30-60fps
AE, AWB, focus locked
Respiratory rate
front camera, propped
30-60s, flash off
On-device signal
processing
detrend, bandpass,
peaks, FFT
Quality gate passed?
No
Yes
Retry guidance
no streak break, no guilt
copy
Show as trend band,
never a precise digit
Tag with cycle phase
Compare to her own
phase-specific baseline
Feed n-of-1 engine

FIGURE 3

ID	REQUIREMENT	PHASE	PRI
FR-301	Fingertip PPG heart rate: rear camera with torch, 30 seconds. Red-channel spatial mean, detrend with smoothness priors at lambda 10, fourth-order Butterworth 0.7 to 3.5 Hz, peak detection cross-checked against FFT. Target accuracy MAPE at or below 5%	MVP	P0
FR-302	Lock auto-exposure, auto-white-balance and focus before capture begins. Exposure hunting is the primary cause of signal loss	MVP	P0
FR-303	Render a live pulse waveform during capture	MVP	P1
FR-304	Display heart rate as a range with confidence, never a single precise digit	MVP	P0
FR-305	Reject captures failing saturation, motion or AC/DC ratio checks. Rejection SHALL be presented as normal, with guidance on what to change, and SHALL NOT break streaks	MVP	P0
FR-306	Contactless respiratory rate: front camera, chest and shoulder region, optical flow bandpassed 0.1 to 0.7 Hz. Target MAE at or below 2 breaths per minute	v1	P1
FR-307	Onboarding SHALL instruct the user to prop the phone for respiratory capture, since handheld capture is the main failure mode	v1	P1
FR-308	Expose HRV from fingertip PPG only as a personal-baseline z-score, never as an absolute value, and never from face rPPG	v1	P1
FR-309	Capture at the highest available frame rate, preferring 60 or 120fps, with peak interpolation for HRV	v1	P1
FR-310	Face rPPG using MediaPipe regions and POS or CHROM projection, labelled experimental. SHALL NOT be the value symptoms are correlated against	v2	P2
FR-311	Gate camera features on device capability at install, with a fully functional non-camera path	MVP	P0
FR-312	Handle expected failure rates gracefully. Between 25% and 45% of face rPPG attempts will fail for Monk Skin Tone 5 to 8 users under warm indoor lighting	v2	P1

9.4 The n-of-1 engine, FR-400

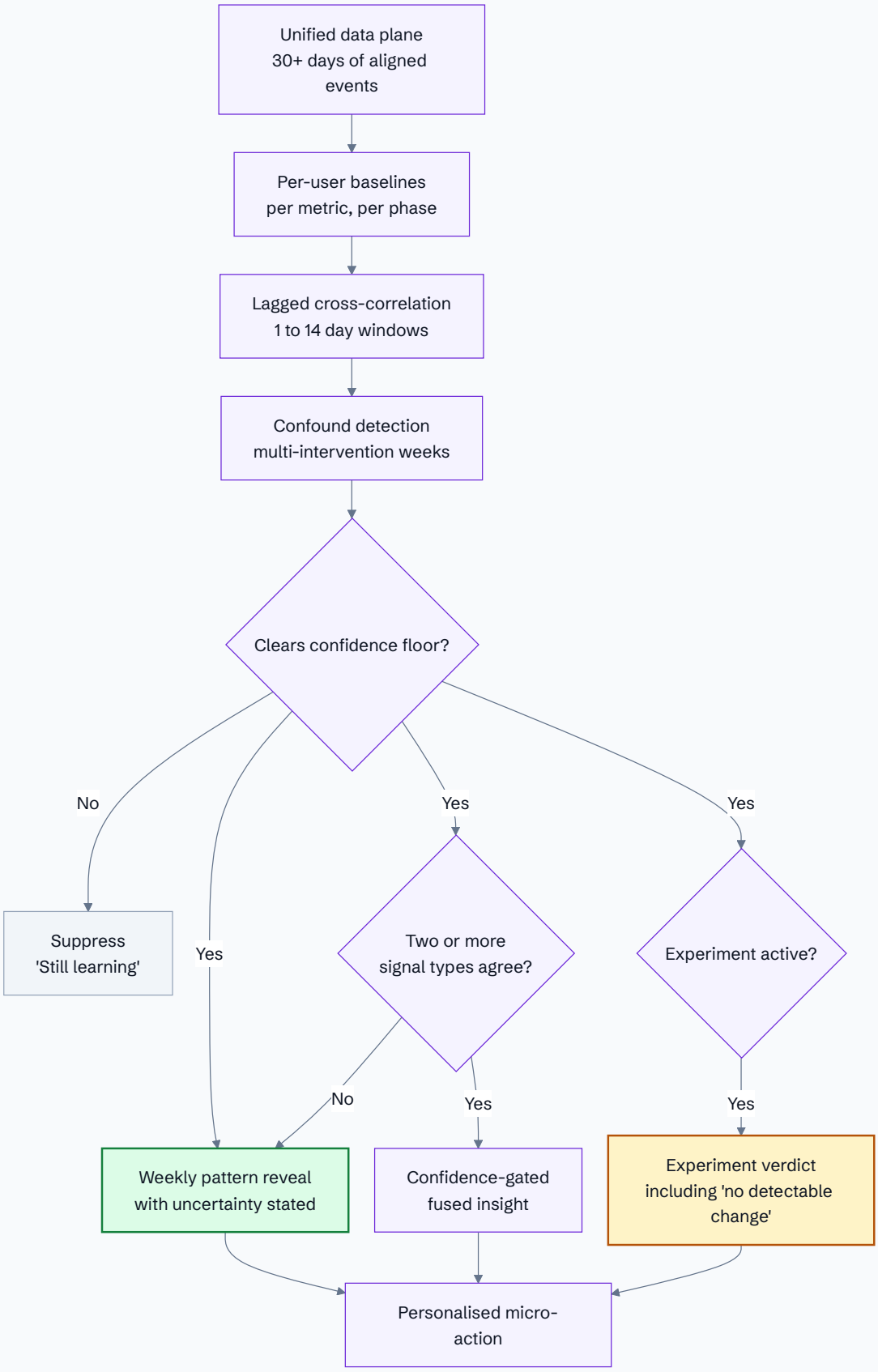


FIGURE 4

ID	REQUIREMENT	PHASE	PRI
FR-401	Compute per-user baselines for every metric, segmented by cycle phase. SHALL NOT compare against population norms	MVP	P0
FR-402	Discover lagged associations across 1 to 14 day windows: sleep to skin, supplement to mood, food to cycle	MVP	P0
FR-403	Apply a confidence floor. Suppress any pattern below threshold and display "still learning". SHALL NOT surface a pattern from insufficient observations	MVP	P0
FR-404	State observation count and signal strength in plain language on every reveal, for example "12 observations, weak signal"	MVP	P0
FR-405	Deliver at least one pattern reveal per active user per week once sufficient data exists	MVP	P0
FR-406	Provide Experiment Mode: pre-declared outcome, baseline period, 8 to 12 week window, washout, and a verdict. "No detectable change" is a valid first-class outcome	v1	P0
FR-407	Ship experiment templates for myo-inositol, sleep window, spearmint, yoga, low-glycaemic eating and metformin adherence	v1	P1
FR-408	Use pre-intervention weeks automatically as the baseline, with no manual setup	v1	P1
FR-409	Warn when multiple interventions change within one window	v1	P1
FR-410	Surface a fused insight only when two or more independent signal types agree, with thresholds tuned for the phone-only signal set	v1	P1
FR-411	Provide aggregate-only, opt-in, k-anonymised cohort priors	Later	P2

9.5 Doctor advocacy, FR-500

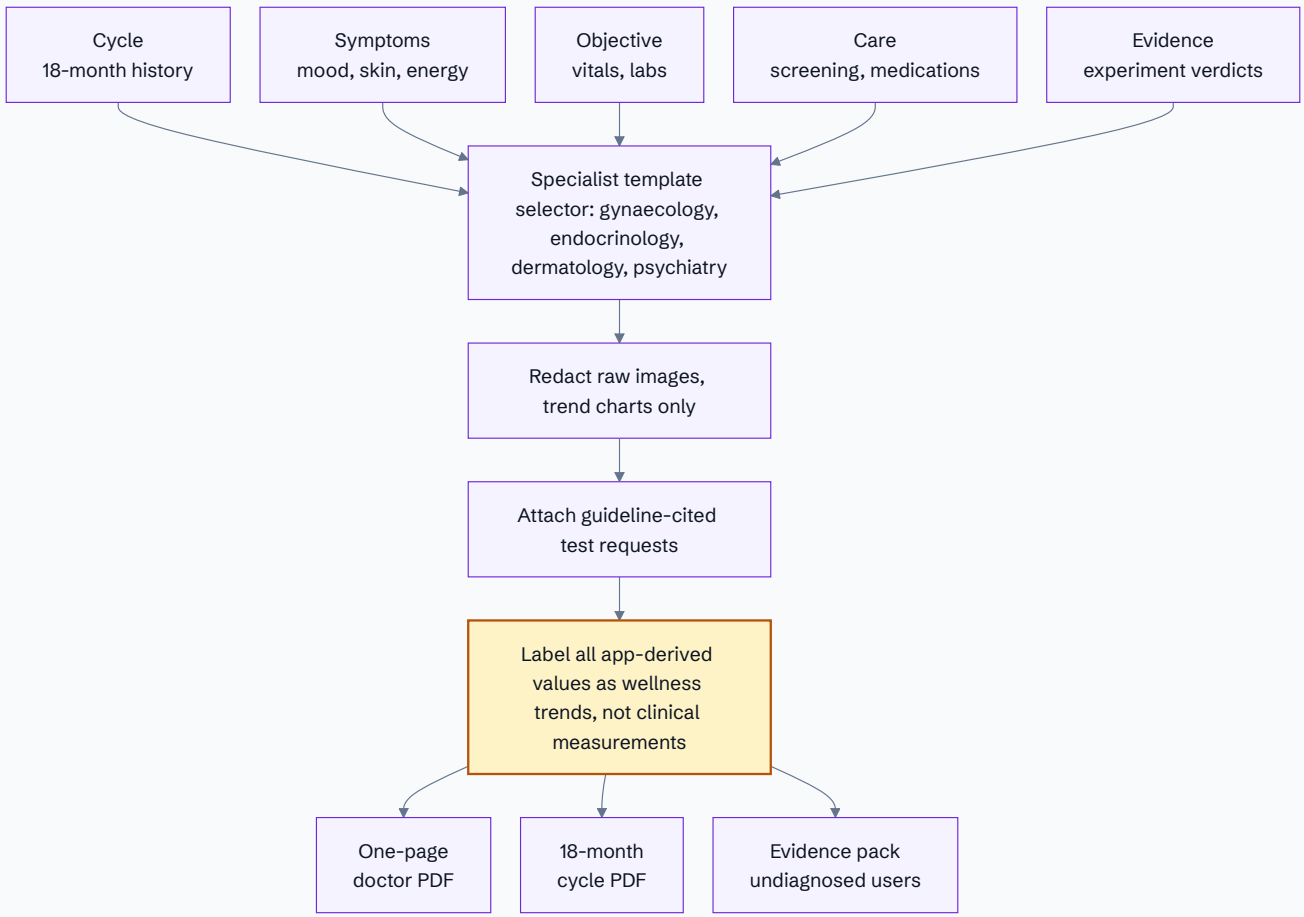


FIGURE 5

ID	REQUIREMENT	PHASE	PRI
FR-501	Provide a one-tap gynaecology export: 18-month cycle history, interval statistics and bleeding log, as PDF	MVP	P0
FR-502	Provide a doctor one-pager: symptom timeline, cycle history, medication log and guideline-cited test requests	MVP	P0
FR-503	Name the tests she can request: fasting insulin, free testosterone, DHEAS, 75g OGTT, AMH	MVP	P0
FR-504	Label all app-derived physiological values in exports explicitly as wellness trends, not clinical measurements	MVP	P0
FR-505	Include trend charts only in exports, never raw photographs	v1	P0
FR-506	Provide specialist-specific templates for gynaecology, endocrinology, dermatology and psychiatry	v1	P1
FR-507	Provide evidence-pack mode for undiagnosed users: symptom evidence, diagnostic criteria explainer and question scripts. Rename from the earlier label, per section 5.3	v1	P1
FR-508	Provide guideline-backed question scripts for asking a clinician directly	v1	P0

ID	REQUIREMENT	PHASE	PRI
FR-509	Provide localised referral pathways so "see a doctor" names a clinic she can reach, by city and budget	v1	P1

9.6 Mental health and safety, FR-600

One mishandled crisis ends the product. This module carries zero tolerance.

ID	REQUIREMENT	PHASE	PRI
FR-601	Run PHQ-4 and GAD-2 check-ins on a defined cadence	MVP	P0
FR-602	Pre-script and human-review the crisis flow. Generated text SHALL NOT appear on any crisis path under any circumstance	MVP	P0
FR-603	Crisis resources, Tele-MANAS 14416, SHALL NEVER be paywalled, login-gated or rate-limited	MVP	P0
FR-604	Gate food-tracking depth behind an eating-disorder screen. On a positive screen, suppress depth and surface support	MVP	P0
FR-605	Provide full PHQ-9 and GAD-7 on guideline cadence, using formally validated Hindi and Tamil translations	v1	P0
FR-606	Suppress action-heavy prompts on low mental-health days	v1	P1
FR-607	Frame CBT and ACT micro-skills explicitly as skills practice, not therapy	v1	P1
FR-608	Provide a one-tap self-reported stress rating, replacing the removed facial inference feature	v1	P1
FR-609	Escalate immediately on chest pain, fainting, drowsy driving, severe bleeding and suicidality, with localised guidance	MVP	P0

9.7 Coaching and treatment, FR-700

ID	REQUIREMENT	PHASE	PRI
FR-701	Deliver one or two evidence-graded, phase-aware micro-actions daily from a curated set	MVP	P0
FR-702	Display a visible evidence grade and date on every health claim	MVP	P0
FR-703	Use additive, cuisine-native food language. SHALL NOT include calorie counting or macro targets by default	MVP	P1
FR-704	Track movement as weekly minutes, 150 to 300 plus two strength sessions, counting 10-minute entries. SHALL NOT display calories burned	MVP	P1
FR-705	Place weight entry behind an opt-in permission prompt, per the 2023 Guideline. SHALL NOT make it a default or headline metric	MVP	P0
FR-706	Derive personalised coaching from her own n-of-1 patterns	v1	P0

ID	REQUIREMENT	PHASE	PRI
FR-707	Provide OSA screen-and-refer with an urgent drowsy-driving flag. SHALL use unlicensed symptom questions unless ESS or STOP-BANG is properly licensed, per REG-004	v1	P0
FR-708	Provide an evidence-graded supplement library covering purpose, studied dose, certainty grade, and Indian availability and price. Zero commerce, zero affiliate links	v1	P0
FR-709	Provide a prescribed-treatment companion: education, adherence and side-effect logging, prescriber question lists. SHALL NOT advise starting, stopping or dosing any prescription	v1	P1
FR-710	Provide GLP-1 safety education covering restored ovulation and contraception, neutral and non-promotional	v1	P0
FR-711	Provide a yoga and breathwork library as a logged intervention with honest evidence framing	v1	P1

9.8 Labs and screening, FR-800

ID	REQUIREMENT	PHASE	PRI
FR-801	Provide a guideline screening ledger: blood pressure annually, lipids at diagnosis, glycaemic status every one to three years, OGTT pre-pregnancy, tracked as due, done and trended	v1	P0
FR-802	Provide a lab vault for fasting glucose, insulin, HOMA-IR, LH to FSH ratio, androgens, lipids, thyroid, vitamin D and HbA1c, with dates	v1	P1
FR-803	Extract values from a photographed printed lab report on device, presented for user confirmation. SHALL NOT auto-save unconfirmed values	v2	P1
FR-804	No feature SHALL be gated behind a laboratory value. Every downstream capability degrades gracefully to zero labs	v1	P0
FR-805	Provide OGTT reminders, preparation guidance and a result-logging template. This is the guideline-recommended path and our substitute for CGM	v1	P1
FR-806	Provide vitamin D test-then-correct guidance: push the test, never the supplement, with the clinician setting dose	v1	P1
FR-807	Label blood-pressure entries as measured at a clinic or pharmacy, since we do not measure blood pressure	v1	P0

9.9 Skin and visual tracking, FR-900

ID	REQUIREMENT	PHASE	PRI
FR-901	Provide guided selfie capture with a fixed face outline and a lighting check that refuses poor shots	v1	P1
FR-902	Provide a self-relative time-lapse the user interprets herself. This is the retention feature and requires no model	v1	P1

ID	REQUIREMENT	PHASE	PRI
FR-903	The system SHALL NOT display beauty, attractiveness or symmetry scores, SHALL NOT enable cross-user comparison, and SHALL NOT produce before-and-after collages	MVP	P0
FR-904	Require separate unbundled consent for skin images, with one-tap deletion and no training use without explicit opt-in	v1	P0
FR-905	Run skin scoring on device. Images SHALL NOT leave the device in production	v2	P0
FR-906	Provide a coarse four-level acne category only if fine-tuned on a purpose dataset. Lesion counts SHALL NOT be displayed, since detection mAP of roughly 0.54 makes them unreliable	v2	P2

9.10 Sleep, FR-1000

ID	REQUIREMENT	PHASE	PRI
FR-1001	Track sleep regularity by bed and wake consistency, using timing rather than duration alone	v1	P1
FR-1002	Derive a sleep window and restlessness index from the accelerometer. This runs in the background and reports gross timing, not sleep stages	v2	P2
FR-1003	Detect snoring from on-device audio during a user-started session. Raw audio SHALL be analysed on device and discarded, never uploaded	v2	P2
FR-1004	Require explicit user initiation for sleep sessions, display the platform recording indicator, and handle OEM-truncated nights gracefully	v2	P2

9.11 AI companion, FR-1100

ID	REQUIREMENT	PHASE	PRI
FR-1101	Answer questions from a curated guideline corpus using retrieval, with visible source citations	MVP	P1
FR-1102	The companion SHALL NOT diagnose, estimate condition likelihood, or recommend or adjust any dose	MVP	P0
FR-1103	Run an on-device model where hardware supports it, with cloud retrieval as fallback. Model location SHALL be disclosed to the user	v2	P1
FR-1104	Route out-of-scope questions to "discuss with your doctor" rather than answering them	MVP	P0

9.12 Privacy, trust and platform, FR-1200

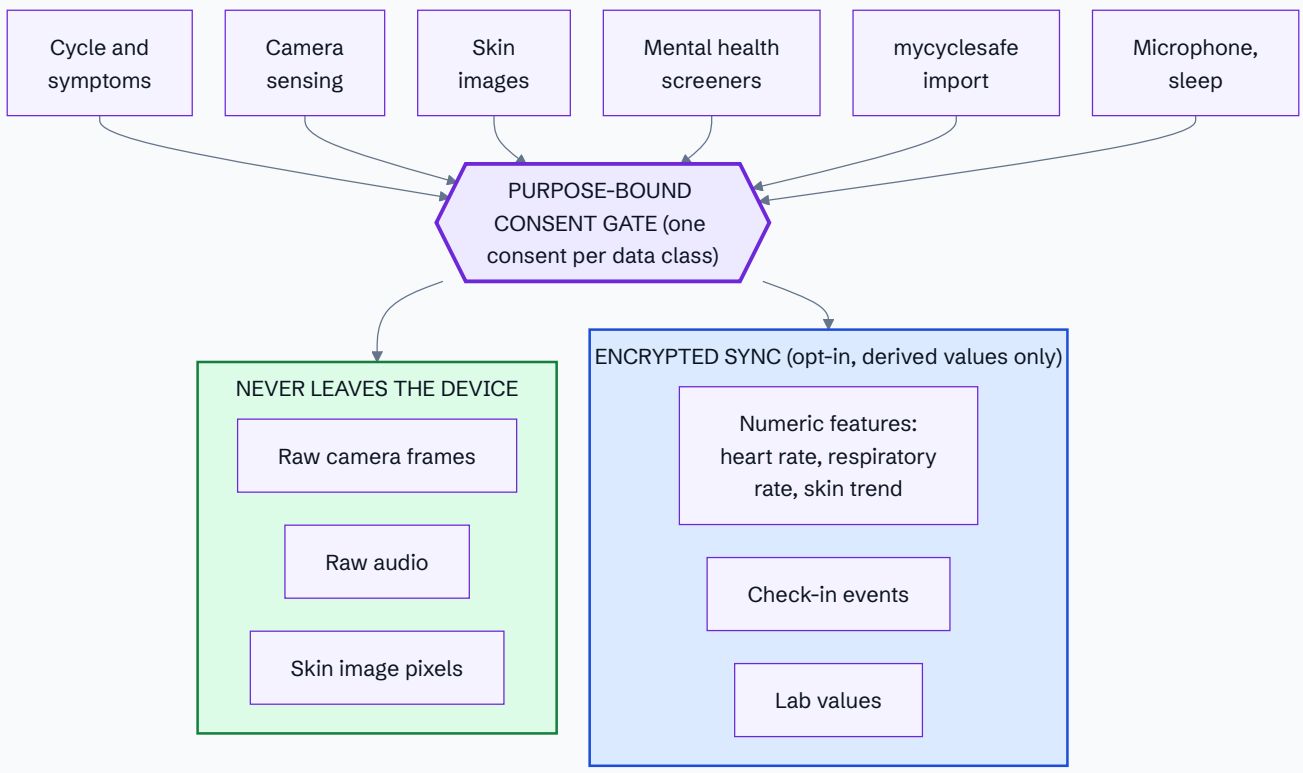


FIGURE 6

Raw media is processed on the device and discarded. Only derived numbers ever sync, and only with consent. Four things are never collected under any configuration: ad-tech SDKs, third-party analytics on health events, cross-user identifiable data, and raw video or audio.

ID	REQUIREMENT	PHASE	PRI
FR-1201	Collect purpose-bound granular consent per data class. Consent SHALL NOT be inherited from mycyclesafe	MVP	P0
FR-1202	Provide one-tap full export and full deletion	MVP	P0
FR-1203	The system SHALL NOT integrate ad-tech SDKs or third-party analytics on health events, and SHALL NOT sell data	MVP	P0
FR-1204	Provide a consented cycle-history import from mycyclesafe to remove cold-start	MVP	P0
FR-1205	Provide a user-configurable biometric application lock	v1	P1
FR-1206	Write back to Health Connect and HealthKit so her data remains portable	v2	P2
FR-1207	Raw video, audio and images SHALL NOT be uploaded under any configuration	MVP	P0
FR-1208	Provide a local-first mode with full offline operation and no account	v2	P2

9.13 Notifications, FR-1300

ID	REQUIREMENT	PHASE	PRI
FR-1301	Every notification SHALL carry a personal insight or a due action. Generic engagement prompts SHALL NOT be sent	MVP	P0
FR-1302	Send at most one or two notifications a day, with user-controlled quiet hours	MVP	P0
FR-1303	The system SHALL NOT use guilt-based streaks or punitive gamification	MVP	P0
FR-1304	Write re-engagement copy that expects lapses	v1	P1

10. Non-functional requirements

10.1 Performance, NFR-100

ID	REQUIREMENT	TARGET
NFR-101	Application cold start	2.5s or less on a ₹10,000-class Android
NFR-102	Daily check-in completion	30s median
NFR-103	PPG result after capture ends	3s or less
NFR-104	Pattern-reveal computation	5s or less over 18 months of data
NFR-105	Doctor PDF generation	10s or less
NFR-106	Application bundle size	60 MB or less initial download
NFR-107	Battery drain per 30s capture	1% or less

10.2 Reliability, NFR-200

ID	REQUIREMENT	TARGET
NFR-201	Crash-free sessions	99.5% or better
NFR-202	Data loss on unexpected termination	Zero. Check-ins persist atomically
NFR-203	Offline capability	Full logging, viewing and insight display
NFR-204	Sync conflict resolution	Last-write-wins per field, with audit trail
NFR-205	Graceful degradation	Every feature functions with zero sensor data

ID	REQUIREMENT	TARGET
NFR-206	OEM battery-killer resilience	Truncated background sessions handled without user-visible error

10.3 Security, NFR-300

ID	REQUIREMENT
NFR-301	Encrypt data at rest with AES-256, keys held in Secure Enclave or Android Keystore
NFR-302	Use TLS 1.3 or later with certificate pinning
NFR-303	Offer an optional biometric application lock
NFR-304	Exclude health data from logs, crash reports and analytics payloads
NFR-305	Never persist raw media beyond the processing window
NFR-306	Commission an annual third-party penetration test before any scale release

10.4 Privacy, NFR-400

ID	REQUIREMENT
NFR-401	Collect only what a declared feature consumes
NFR-402	Process all camera, audio and image data on device
NFR-403	Provide granular per-class consent with independent revocation
NFR-404	Provide full export in a machine-readable format
NFR-405	Complete deletion within 72 hours across all systems
NFR-406	Publish a plain-language privacy policy at eighth-grade reading level or below

10.5 Accessibility, NFR-500

ID	REQUIREMENT
NFR-501	Meet WCAG 2.1 AA across all user-facing surfaces
NFR-502	Support TalkBack and VoiceOver fully
NFR-503	Maintain 4.5:1 contrast minimum, never using colour alone to convey state
NFR-504	Support dynamic type to 200% without layout breakage
NFR-505	Provide touch targets of 44 by 44 points or larger

ID	REQUIREMENT
NFR-506	Provide a non-visual guidance path for camera capture

10.6 Localisation, NFR-600

ID	REQUIREMENT
NFR-601	Ship English at launch, Hindi and Tamil at v1
NFR-602	Use formally validated translations for clinical screeners. Machine translation is prohibited
NFR-603	Adapt cuisine and coaching content regionally rather than translating it
NFR-604	Format dates, numbers and units per locale

10.7 Content and safety, NFR-700

ID	REQUIREMENT
NFR-701	Attach an evidence grade and review date to every health claim
NFR-702	Have a qualified clinician review content before release
NFR-703	Prohibit cure, reverse, restore-fertility and guaranteed-outcome language everywhere, including influencer and cross-promotional copy
NFR-704	Have a mental-health professional review crisis content
NFR-705	Re-review the evidence base on guideline updates: NICE December 2026, international refresh 2028

10.8 Compatibility, NFR-800

ID	REQUIREMENT
NFR-801	Support Android 8.0 and later, the Health Connect floor, and iOS 14 and later
NFR-802	Function on 2 GB RAM devices
NFR-803	Gate camera features on a capability check with a functional fallback
NFR-804	Design portrait-first, tablet-tolerant

10.9 Observability, NFR-900

ID	REQUIREMENT
NFR-901	Exclude health-data values from product analytics. Events and counts only
NFR-902	Monitor PPG capture success rate segmented by device class and skin tone
NFR-903	Log crisis-path invocations as counts only, reviewed weekly
NFR-904	Monitor pattern-reveal suppression rate to detect an over-tight confidence floor

11. Regulatory and compliance

11.1 Age

REG-001: the system SHALL enforce an 18+ age gate and SHALL NOT process data for users under 18.

India's DPDP Act classifies anyone under 18 as a child. Processing requires verifiable parental consent, verifying both the child's age and the parent's identity, and the Act expressly prohibits tracking, behavioural monitoring, targeted advertising and profiling of children.

The Rule 11 healthcare exemption covers a clinical establishment, mental health establishment or healthcare professional, restricted to providing health services. We cannot claim it while also claiming non-clinical status for CDSCO, because the two positions are mutually exclusive.

The deeper obstacle is that our n-of-1 engine performs profiling. Lagged correlation over longitudinal behavioural data is precisely the prohibited activity, so no consent checkbox resolves it.

This costs us something real. PMOS commonly presents in adolescence, adolescent sleep-apnoea prevalence runs at 29%, and our own research shows teenagers need help immediately. We exclude them for legal reasons, and adolescent support becomes a funded roadmap item requiring verifiable-parental-consent infrastructure.

11.2 Medical device boundary

REG-002: the system SHALL operate as screen-and-refer. It SHALL NOT diagnose, produce condition-likelihood scores, or recommend or adjust doses.

REG-003: the system SHALL NOT measure or display blood pressure, blood oxygen, blood glucose or body temperature from device sensors.

App Store Review Guideline 1.4.1 states that apps claiming to measure blood pressure, body temperature, blood glucose levels or blood oxygen levels using only the sensors on the device are not permitted. Renaming an output to "BP trend" or "cardiovascular load" does not avoid classification.

CDSCO has issued final guidance on medical device software, with risk-based Class A to D and wellness software held to a lower standard than diagnostic software. Class A non-sterile non-measuring devices are exempt from full licensing under G.S.R. 777(E), though registration and post-market surveillance still apply, and ISO 13485 is a practical prerequisite for market access.

11.3 Data protection

ID	REQUIREMENT
REG-005	Collect DPDP-compliant purpose-bound consent, with no inheritance from mycyclesafe
REG-006	Honour granular deletion and export rights within 72 hours
REG-007	Sell no data, integrate no ad-tech SDKs, run no third-party health analytics

11.4 Clinical instrument licensing

REG-004: the system SHALL NOT reproduce or score a copyrighted instrument without a valid licence.

INSTRUMENT	STATUS	ACTION
PHQ-9, GAD-7, PHQ-4	Free. Pfizer released them in 2010 with no copyright restriction. Commercial use and translation permitted	Use freely. Confirm which Hindi and Tamil translations are formally validated
Epworth Sleepiness Scale	Paid commercial licence via Mapi Research Trust	License or avoid
STOP-BANG	CC BY-NC-ND. Non-commercial only, no derivatives without permission	Obtain written permission or avoid
Berlin Questionnaire	Unclear	Verify before use

The recommended path is unbranded symptom questions covering snoring, unrefreshing sleep, daytime sleepiness and witnessed apnoea, which prompt a doctor visit without reproducing or scoring a copyrighted scale.

11.5 Store compliance

ID	REQUIREMENT
REG-008	Complete the Google Play health apps declaration. Budget two to four weeks review latency and possible rejection
REG-009	Disclose device compatibility in the store listing where sensor-based health functions are used
REG-010	Include an in-app "not a medical device" disclaimer and clinician-consultation reminder
REG-011	Disclose accuracy methodology for any accuracy claim. SHALL NOT cite another company's figures as ours
REG-012	Prepare an App Review 2.5.4 and 2.5.14 justification before shipping overnight audio

11.6 Marketing claims

ID	REQUIREMENT
REG-013	Prohibit cure, reverse and restore-fertility claims across the entire chain: application, store listing, mycyclesafe cross-promotion and influencer content. The DMR Act 1954, FSSAI and ASCI all reach this material
REG-014	Attach a visible evidence grade to every content claim

12. Release plan

PHASE	WINDOW	WORKSTREAMS
MVP	Aug – Oct 2026	Cycle engine and check-in. n-of-1 engine v0. **Fingertip PPG** (critical path). Doctor exports and crisis floor
v1	Oct 2026 – Mar 2027	**Experiment Mode** (critical path). Skin trend, labs, screening ledger. Respiratory rate and HRV. Hindi and Tamil
v2	Mar – Jul 2027	On-device LLM companion. Sleep actigraphy and snore detection. Lab OCR and face rPPG beta
Later	Aug 2027 onward	Cohort priors. Clinician portal

12.1 MVP definition of done

The irregular-native cycle engine with interval sentinel. The 30-second check-in with pause mode. The n-of-1 pattern reveal with confidence gating. Fingertip PPG heart rate with quality gate. Gynaecology export, doctor one-pager and test-request helper. The crisis-safe mental-health floor. Dignity defaults covering weight opt-in, the 18+ gate and anti-beauty guardrails. The mycyclesafe bridge with granular consent. Evidence-graded education with the retrieval companion.

12.2 Hackathon demonstration

Five beats, in order.

1. Open with the rename. Two months ago medicine renamed this condition to acknowledge it was never about cysts or fertility, and every product still treats it as though it were.
2. Fingertip PPG live. A judge places a finger on the camera and the pulse waveform draws itself in real time.
3. The reveal. Your skin flares two to three days after nights under six hours. Not your face wash, your sleep.
4. The export. One tap produces an 18-month doctor-ready PDF.
5. Close on distribution. A ₹10,000 phone, no watch, no strips, nothing to buy. One in eight women, and we already reach 500,000 of them.

13. Risks

#	RISK	SEVERITY	MITIGATION
R1	Competitors already ship features we claimed. PMOS Vera is live with irregular-native framing, exports, weight-neutral design and a photo journal	Critical	Lead with Experiment Mode, camera vitals and India-native depth. Rewrite the near-verbatim copy immediately
R2	Our privacy claim is weaker than Vera's, which runs no servers	High	Ship on-device inference per ADR-004. Reframe as on-device for sensing, encrypted sync for the rest. Do not lead with privacy
R3	The PPG quality gate rejects too often, particularly for Monk Skin Tone 5 to 8	High	Fingertip PPG, which is skin-tone-fair, remains primary. Face rPPG stays in beta. Failure is treated as normal, with a lighting pre-check and monitoring by skin tone per NFR-902
R4	Small-sample correlation surfaces a false pattern	Critical	Hard confidence floor, observation count always stated, "still learning" as a first-class state
R5	Skin capture worsens body-image distress	Critical	Self-relative only, no scores, opt-in, eating-disorder screen, distress reports tracked as a counter-metric
R6	A crisis is mishandled	Critical	No generated text on crisis paths, human-reviewed scripts, never paywalled, clinician-reviewed
R7	Store rejection over health declaration, camera justification or accuracy claims	High	Complete declarations before submission, disclose methodology, never quote another company's accuracy figures
R8	Losing the adolescent segment to DPDP	High	18+ at launch, with verifiable-parental-consent infrastructure funded as a roadmap item
R9	Retention collapse, the usual fate of consumer health apps	Critical	The pattern reveal is the retention mechanism. Lapse-tolerant design, no guilt
R10	OSA instrument licensing blocks a P0 feature	High	Ship unbranded symptom questions per REG-004
R11	Indian OEM battery managers truncate background sessions	Medium	Onboarding walkthrough to disable optimisation, graceful truncation handling
R12	The mycyclesafe bridge underperforms	High	Treat conversion as an assumption to test in week one, not a certainty, and hold an organic acquisition plan in reserve

14. Open decisions

#	DECISION	OPTIONS	RECOMMENDATION
D1	Standalone application or a module inside mycyclesafe	Standalone, embedded	Standalone with a deep bridge. Cleaner consent, PMOS-native identity, app-store search presence
D2	Pricing and the freemium line	Not set	Anchor at ₹200 to ₹500 monthly. Safety paths always free. Monetise personalisation, never safety
D3	Product name	Cyster, PMOS-forward, condition-neutral	Condition-neutral scales to other conditions later. PMOS-forward wins search. Unresolved
D4	How hard to lead with "PMOS"	Not set	Dual-name for three years and own both search surfaces
D5	Clinician portal timing	Year one, or after NICE publishes	Defer until after December 2026 for format alignment
D6	Hindi and Tamil together at v1, or Hindi first	Not set	Unresolved. Depends where mycyclesafe's Indian users concentrate

15. Appendix

15.1 Glossary

TERM	MEANING
PMOS	Polyendocrine Metabolic Ovarian Syndrome, renamed from PCOS by a Monash-led global consensus published in The Lancet, 12 May 2026
n-of-1	A single-subject experiment. Here, finding cause and effect within one user's own data
rPPG	Remote photoplethysmography, extracting a pulse signal from video
Contact PPG	Fingertip on camera with torch. Higher signal-to-noise, and skin-tone-fair
MST	Monk Skin Tone scale, 1 to 10. India sits predominantly at 5 to 8
SaMD	Software as a Medical Device
DPDP	Digital Personal Data Protection Act 2023, India
mFG	Modified Ferriman-Gallwey hirsutism scale
OGTT	Oral glucose tolerance test, the guideline-recommended glycaemic assessment

TERM	MEANING
Confidence gate	The threshold below which a pattern is suppressed as "still learning"

15.2 Sources

Clinical. The 2023 International Evidence-based Guideline. The PMOS rename. NICE GID-NG10436, with consultation running 1 July to 11 August 2026, publication expected 9 December 2026, already using "PMOS". Sleep apnoea prevalence of 37% against 6%, odds ratio 9.52, in Frontiers in Endocrinology.

Sensing. Google's camera heart rate at MAPE 1.6% and respiratory rate at MAE 0.78, in Communications Medicine. Google PHRM in Nature 2026, cited for the science only, since it is not an available capability. rPPG algorithm evaluation in npj Digital Medicine. PPG sampling requirements for HRV. Barrett et al. 2019 on facial expression and emotion inference.

Platform and regulatory. App Store Review Guidelines, section 1.4.1. AVCaptureSession background interruption. Android foreground service types. Health Connect access declaration. DPDP child data under Rule 10 and the Rule 11 exemption scope. CDSCO medical device software guidance. PHQ and GAD free access. Epworth licensing.

Technical. The Capacitor health plugin.

Market. India wearables decline 2025.

15.3 Document history

VERSION	DATE	CHANGE
1.0	July 2026	Consolidated PRD, superseding documents 01 to 03. Incorporates the feasibility audit: passive sensing removed, external devices removed, the 18+ decision, the competitor correction, and the hybrid architecture